MATERIALS AND METHODS

Numerical Model Setup

Hydrodynamic circulation in the MGLS was simulated using the software SisBaHiA - Sistema Base de Hidrodinâmica Ambiental, a process-based modeling system enhanced and supported by the Federal University of Rio de Janeiro, developed for application in studies of free-surface water bodies (SILVA et al., 2020; ELANE et al., 2022; SARANA CASTRO DEMONER et al., 2023; A. TROG-FERREIRA et al., 2024; SAWAKUCHI et al., 2017; PEIXOTO; ROSMAN; VINZON, 2017; MARIA; PAULO, 2018; IAEA, 2019; REIS & SILVA, 2024; ROVERSI; ROSMAN; HARARI, 2016). Figure 1 shows the finite elements mesh adopted for the lagoon system with the proposed alternatives and points of interest which will be analyzed in this study. Three hypothetically permanent tidal inlets were considered: i) the Barra Channel, which needs to be dredged and stabilized; ii) Dredging, clearance and stabilization of the whole course of the Costa Channel, which currently presents several flow interruptions; iii) Creation of a new tidal inlet in the eastern portion of the Costa Channel, near the Maricá Lagoon.

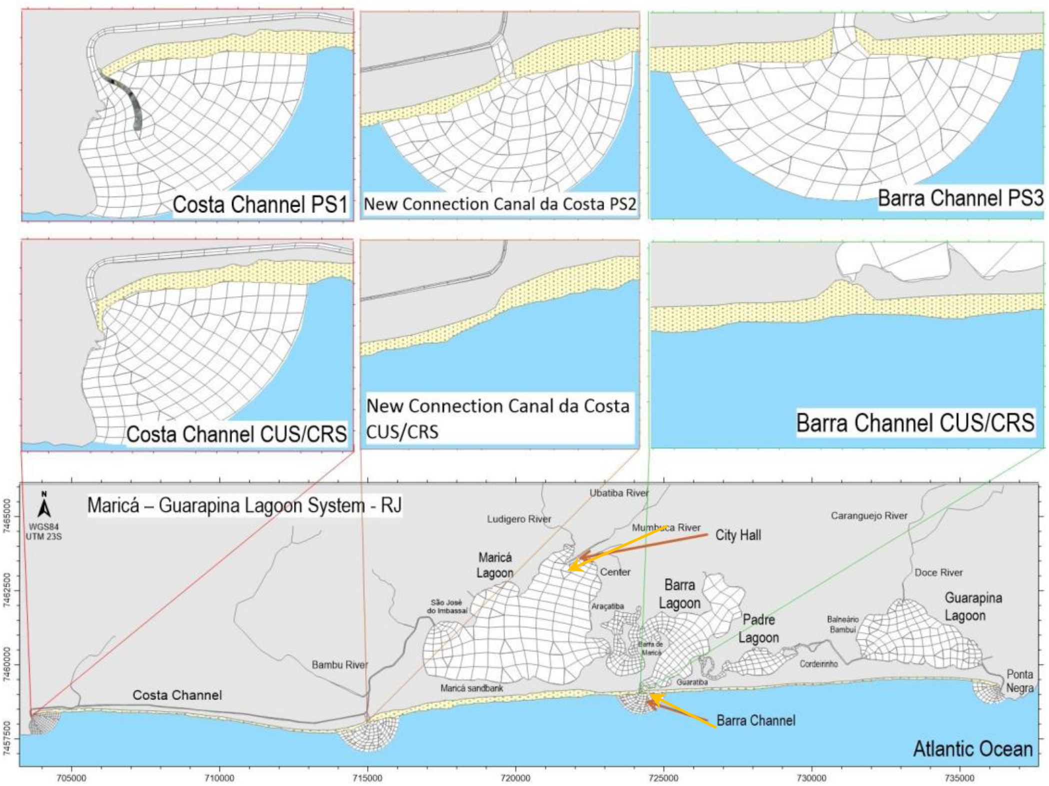

Figure 1: Finite elements mesh adopted for the Maricá-Guarapina Lagoon System, with the proposed

maritime connections.

The models used in each analysis will be detailed further below.

Hydrodynamic Model

The applied 2DH hydrodynamic model solves the shallow waters Navier-Stokes (Eqs. 1 and 2) and the continuity equation (3) for incompressible flow with homogeneous density in a finite element discretization [10].

$$\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}=-g\frac{\partial \zeta}{\partial x}+\frac{1}{\rho_0(\zeta+h)}\left(\frac{\partial[(\zeta+h)\hat{\tau}_{xx}]}{\partial x}+\frac{\partial[(\zeta+h)\hat{\tau}_{xy}]}{\partial y}\right)+\frac{1}{\rho_0(\zeta+h)}\left(\tau_x^S-\tau_x^B\right) \qquad (1)$$

$$\frac{\partial v}{\partial t}+u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}=-g\frac{\partial \zeta}{\partial y}+\frac{1}{\rho_0(\zeta+h)}\left(\frac{\partial[(\zeta+h)\hat{\tau}_{yx}]}{\partial x}+\frac{\partial[(\zeta+h)\hat{\tau}_{yy}]}{\partial y}\right)+\frac{1}{\rho_0(\zeta+h)}\left(\tau_y^S-\tau_y^B\right) \qquad (2)$$

$$\frac{\partial(\zeta+h)}{\partial t}+\frac{\partial u(\zeta+h)}{\partial x}+\frac{\partial v(\zeta+h)}{\partial y}=0 \qquad (3)$$

In Eqs. (1) – (3) $u(x, y, t)$ and $v(x, y, t)$ are the 2DH velocity components in x and y, respectively; g is the gravity acceleration; $z = \zeta(x, y, t)$ is the free surface and h is the bathymetry; ρ_0 is the reference water density; $\hat{\tau}_{xx}$, $\hat{\tau}_{xy}$,$\hat{\tau}_{yx}$ and $\hat{\tau}_{yy}$ are the turbulent shear stresses; τ^S and τ^B are the shears stresses on the free surface and on the bottom, respectively.

Eulerian Model

The water age and water renewal simulations were performed with the Eulerian Model (EM) of advective–diffusive transport with kinetic reactions. This model is used to analyze Water Renewal Rates and Water Age, as performed by AGUILERA; LÍGIA; ROSMAN, 2020 and JESSYCA PETRY DALAZEN; LOURDES; SANTOS, 2020.

Equation 5 presents the general formulation of the EM, which is the general advective-diffusive transport equation for scalars in shallow water bodies, where C is the concentration of the substance of interest; q_P, q_E, and q_{Ia} are given values of flow rates per unit area, e.g., [m³/s/m²], corresponding respectively to precipitation, evaporation, and infiltration; H is the water column; R represents the kinetic reactions of production or consumption of a given scalar; and the T terms represent turbulent diffusion coefficients.

$$\frac{\partial C}{\partial t}+U\frac{\partial C}{\partial x}+V\frac{\partial C}{\partial y}=\frac{1}{H}\frac{\partial}{\partial x}\left(T_{xx}\frac{\partial C}{\partial x}+T_{xy}\frac{\partial C}{\partial y}+T_{xt}\frac{\partial C}{\partial t}\right)+\frac{1}{H}\frac{\partial}{\partial y}\left(T_{yx}\frac{\partial C}{\partial x}+T_{yy}\frac{\partial C}{\partial y}+T_{yt}\frac{\partial C}{\partial t}\right)-\frac{C}{H}(q_P-q_E)+\frac{(C_{Ia}-C)}{H}q_{Ia}+\Sigma R \qquad (5)$$

Water Renewal Rate

The Water Renewal Rate analysis applied the EM, using a conservative substance as a tracer, without reactions, sedimentation, or resuspension. In this analysis, at the initial instant, all the water within the modeling domain is assigned with a 0% renewal index of, representing “old water”, while the inflows from the main tributaries are characterized with an index of 100%, representing new water. The spatial evolution of this tracer is determined by advective and diffusive transport processes, resulting in mixing of old and new waters over time. Thus, at each mesh node, the renewal rate expresses the rate between new and old water. For example, a 40% water renewal rate in a given point indicates that 40% of the water at that location is new and 60% corresponds to the remaining older water.

Water Age

The Water Age analysis is also based on the use of a passive substance subject to first-order decay, with an initial concentration of $C_0 = 1$ throughout the domain. The temporal evolution of this substance is described by the following differential equation:

$$\frac{\partial(C)}{\partial t}=-K_dC \;=>\; C(t)=C_0e^{-K_dt} \qquad (4)$$

In equation (4) K_d is the constant decay, related to characteristic half-time (T_{50}) and the order of magnitude (T_{90}). The water age at a given point is defined as:

$$IA(x,y,z)=\frac{ln(\frac{C_0}{C(x,y,z)})}{K_d} \qquad (5)$$

which expresses the mean residence time of the water in the system. Inflows of new water are associated with zero age, while transport and diffusion redistribute the values across the domain. The results can be represented as isoline maps of water age, providing spatial estimates of residence time and allowing the assessment of environmental processes that depend on water retention time, such as eutrophication.

Simulated Scenarios

To evaluate the effects of different engineering interventions on flood mitigation and water renewal, five scenarios were simulated (Table 1): one representing the current configuration of the MGLS (CS — Current Scenario) and four projected alternatives (PS1, PS2, PS3 and PS4) involving the stabilization or creation of tidal channels. Each of these five scenarios was simulated under two forcing conditions: Extreme Rain, corresponding to the 29 February 2016 rainfall event, used to evaluate flood response; and Usual Conditions, representative of normal summer and winter hydrodynamics, used to evaluate water renewal and water age.

In other words, the same five engineering configurations (CS, PS1, PS2, PS3 and PS4) were each simulated under both forcing conditions, so that the same alternatives could be compared consistently for both flood mitigation and water renewal.

Scenario	**Rainfall condition**	**Intervention / configuration**	**Purpose**
CS — Current Situation	29 Feb 2016 extreme rainfall event / Normal conditions	Present-day lagoon configuration, without structural intervention	Baseline hydrodynamic condition under usual conditions
PS1 — Projected Situation 1	29 Feb 2016 event / or Normal conditions	Costa Channel dredged, unobstructed and stabilized	Assessing the effect of stabilizing the Costa Channel in flood control and water renewal

PS2 — Projected Situation 2	29 Feb 2016 event / or Normal conditions	Additional tidal inlet in the eastern portion of the Costa Channel	Test enhanced drainage through an additional ocean connection
PS3 — Projected Situation 3	29 Feb 2016 event / or Normal conditions	Barra Channel dredged and stabilized	Assessing the effect of stabilizing the Barra Channel in flood control and water renewal
PS4 — Projected Situation 4	29 Feb 2016 event / or Normal conditions	Combination of PS1, PS2 and PS3	Assessing the effect of combined situations in flood control and water renewal

Table 1: Summary of the analysis for each scenario.

Sensitivity test	Base scenario	Tested parameter	Values
Barra Channel depth sensitivity	PS4	Barra Channel depth	0.4, 0.8, 1.4 and 2.0 m

Table 2: Summary of the Sensitivity test.

Environmental Data

Bathymetry

The bathymetric information (Figure 2) was obtained from a data compilation by the Non-Governmental Organization Lagoa Viva (CODEMAR, 2025), the Directorate of Hydrography and Navigation (DHN, 2025) of the Brazilian Navy, and field measurements.

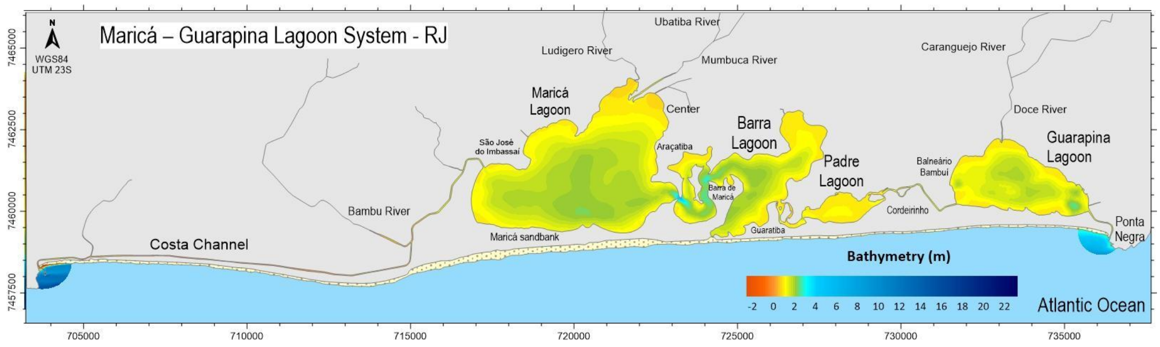

Figure 2: Bathymetry of the Maricá-Guarapina Lagoon System.

On 13 February 2017, a field survey was conducted in the region of the Barra Channel, which is often opened for flood control. An optical level and a leveling staff were used to measure the selected points, which were georeferenced using GPS. Figure 3 illustrates this fieldwork.

Figure 3: Field survey for topo-bathymetric data collection.

The surveyed points delimited the left and right sandbanks of the Barra Channel, using the sea-to-lagoon direction and the shoreline at a given moment (11:15 a.m.) as reference. These points also defined three cross-sections along the channel: the first one following a power line which crosses the channel, representing the area where the dry channel curves toward the lagoon and leaves the dune region; the second one, underneath the bridge which crosses the channel entrance into the lagoon, marking the boundary of the flooded area; and the last one within the lagoon, in a deeper region (around 2 m), representing a permanently submerged area. The surveyed profile points are presented in Figure 4. Additional marginal points were also marked, to represent the extent of the 2016 flood event, which were identified through residents' reports and flood marks on buildings. The leveling procedure revealed that the 2016 flood water level was around 0,93 m above the lagoon mean water level. According to news reports, 24 hours after the beginning of the rain a

1 m water level was measured (“Canal em Maricá não é aberto e água da chuva continua em condomínio,” 2016), which is consistent with the flood water level measured in the field.

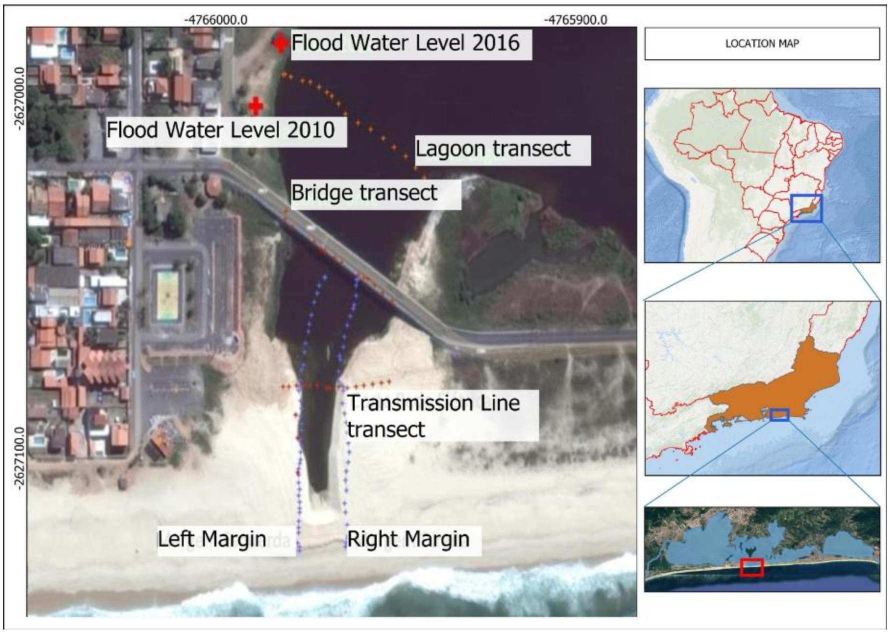

Figure 4: Points marked in the field with GPS and flood water level during the 2010 and 2016 floods.

Tides

The tidal data used in the flood analysis models were obtained through harmonic constituents previously calibrated for the Guanabara Bay region, with the computational grid extending to an area near the Coast Channel (Ramos et al., 2024), considering the period of interest: 15 days starting on 29 February 2016, which is date of the flood. Additionally, storm surge water level was incorporated using data from the time series of the tide gauge located at Ilha Fiscal (RJ), the same used for calibrating the harmonic constituents. Figure 5 illustrates the contributions of the astronomical tide, storm surge and the total water level, which is the sum astronomical tide and storm surge water levels.

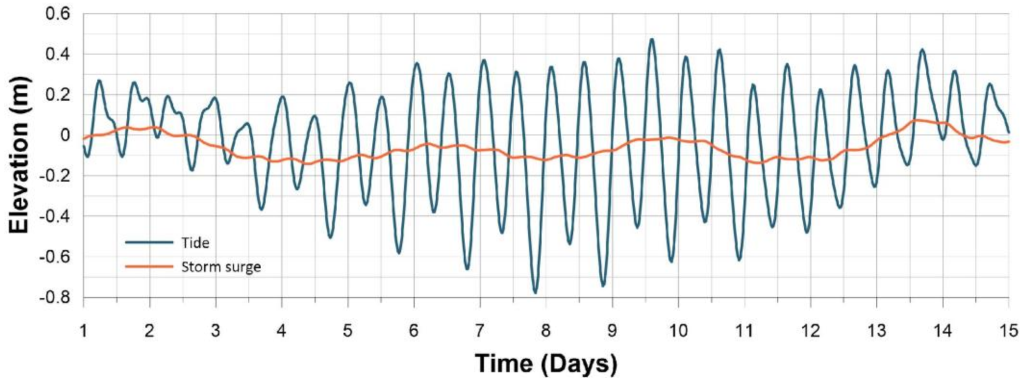

Figure 5: Water level time series for the flood analysis.

For the water renewal analysis, tidal data were obtained from the same harmonic constituents previously mentioned for the periods of January 2024, representing summer, and July 2024, representing winter. Figure 6 illustrates the contributions of astronomical tide, storm surge, and their sum, the total tide, which was used in the model.

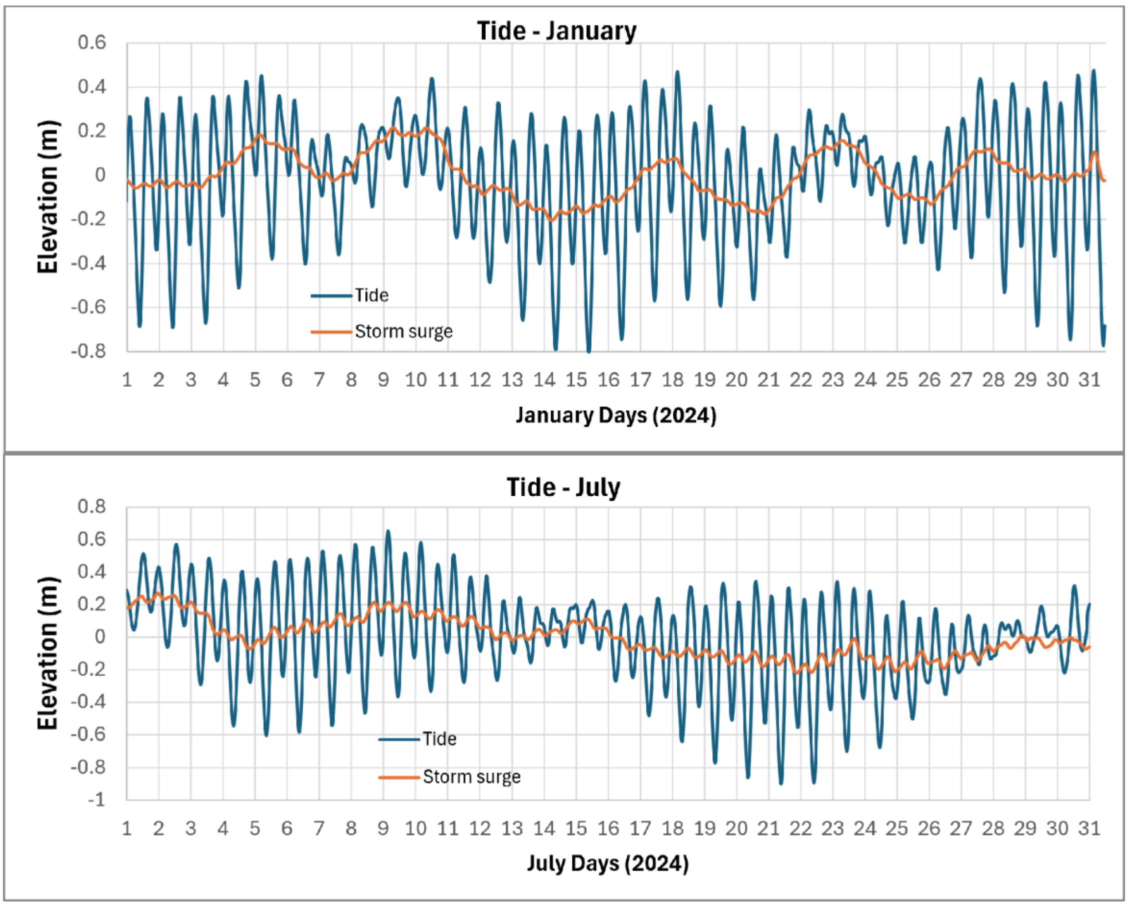

Figure 6: Tidal data from the simulated period in the hydraulic time analysis models.

Winds

The stations were carefully selected to accurately represent the local wind regime through adequate spatial distribution for each analysis proceed (Figure 7).

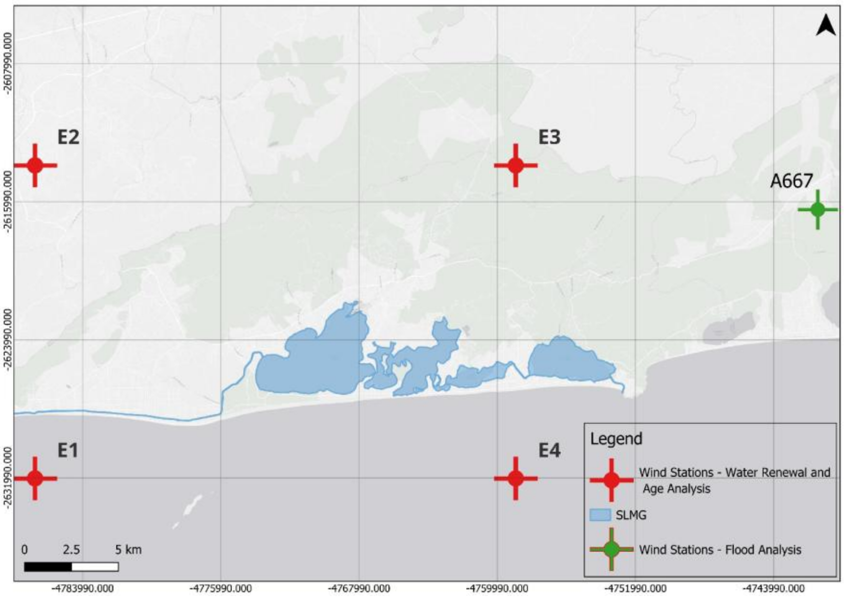

Figure 7: Selected wind seasons.

The 2016 wind data were provided by the National Institute of Meteorology (INMET) (Inmet, 2025) from the A667 automatic weather station located in Saquarema. Figure 8 shows the station's location as well as the wind speed and direction patterns for the region. Given the unavailability of INMET data for 2024, the wind time series applied for the water renewal analysis were obtained from ECMWF (European Centre for Medium-Range Weather Forecasts) global reanalysis models for both scenarios. The Figure 8 shows the wind patterns for those periods.

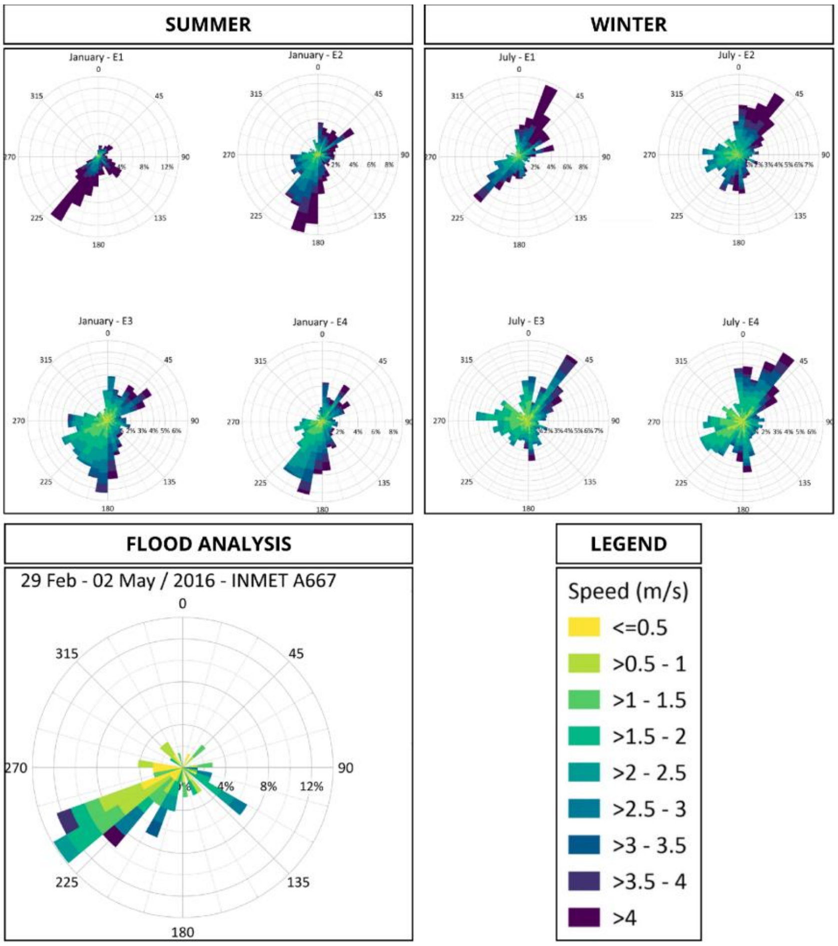

Figure 8: Wind patterns for eulerian and inundation analysis.

Model Calibration (Flow Data)

The MGLS hydrological flow is one of the most critical inputs for flood analysis. Due to the absence of direct measurements of river discharge during this event, the freshwater inflow to the lagoon system was estimated through a rainfall–runoff approach and subsequently adjusted to reproduce the maximum lagoon water levels observed in the field. The total river discharge flowing into the lagoon system was estimated using a rainfall, runoff model based on the Synthetic Unit Hydrograph (UHT), following DNIT (2005). The data for the UHT were obtained through geoprocessing and from the INMET database: Drainage área = 217 km²; Slope = 0.005%; Concentration time = 6 h; Average Curve Number (CN) = 57. From these parameters, rainfall intensity was calculated for each time interval, and the resulting river discharges were obtained through the Unit Hydrograph Technique (UHT). Figure 9 presents the calculated flow hydrograph.

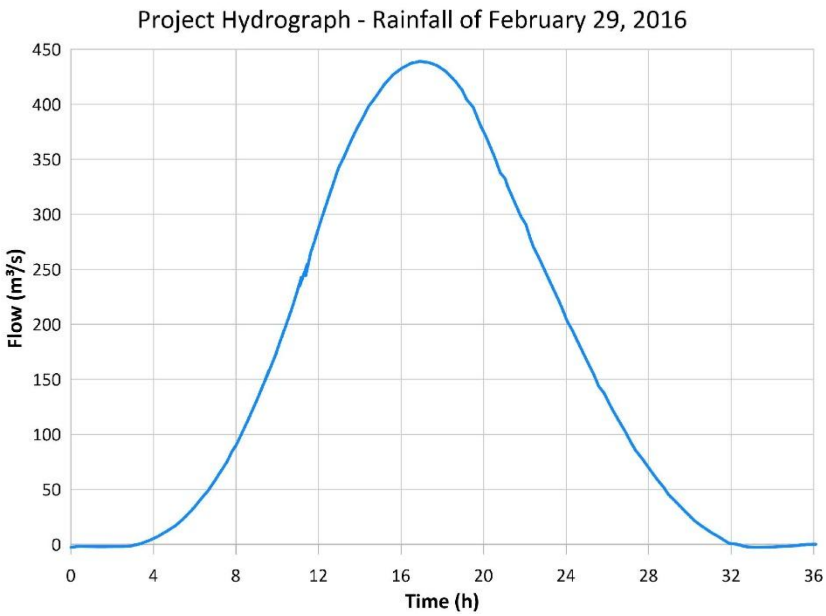

Figure 9: 29 February 2016 calculated rainfall hydrograph.

The resulting river inflow was then used as a starting point for the calibration procedure adopted for the hydrodynamic model, which consisted of slightly adjusting the magnitude of the freshwater inflow hydrograph, so that the simulated lagoon water levels reproduced the maximum water levels observed during the extreme rainfall event of 29 February 2016. Field measurements and observational evidence, including flood marks on buildings and reports from local residents, indicated a 1,0 m maximum water level near the Maricá City Hall and at Barra de Maricá. Since no other measurements are available for that flood event, this indirect measurement is the main data used for model calibration.

The calibration was performed under the Current Situation (CS) scenario, during the extreme event, which represents the lagoon system under the existing configuration, without any structural interventions. During this process, only the magnitude of the freshwater inflow hydrograph was adjusted, while the marine boundary conditions remained unchanged. As shown forward, in Figure 12, the water level in the Maricá Lagoon is very little affected by the tides.